

SAMA ASHRAF ABDELKADER REFAIE

GIS Specialist / GIS Analyst — Remote Sensing & Cartography

Cairo, Egypt | +20 114 038 2034 | samaashraf719@gmail.com | [linkedin.com/in/sama-ashraf](https://www.linkedin.com/in/sama-ashraf) | github.com/sama765 | sama765.github.io/portfolio

PROFILE

Entry-level Geomatics graduate with hands-on academic and field experience across the full GIS workflow — from data collection to final decision-support maps. Specialised in Geographic Information Systems, Remote Sensing and Cartography, with practical experience in multi-criteria analysis (AHP, Weighted Overlay), remote-sensing indices, and publication-ready cartographic output. Detail-oriented, analytical, and a fast learner.

EDUCATION

B.Sc. in Geomatics (Surveying, Mapping, GIS & Remote Sensing) 2026
Faculty of Arts, Cairo University — GPA: 3.13

SELECTED PROJECTS

- Optimal Distribution of EV Charging Stations, Cairo** Graduation Project — Grade: A
 - Spatial analysis of optimal EV/e-motorcycle charging station distribution across Cairo Governorate using multi-criteria analysis (AHP, Weighted Overlay) in ArcGIS Pro.
 - Applied Kernel Density, Nearest Neighbor, Gap Analysis and Network Analyst Service Area to evaluate distribution and road access.
 - Integrated slope, NDVI, NDBI and Land Surface Temperature (SRTM, Sentinel-2, Landsat-8) to rule out unsuitable land and produce a final suitability map.
- LiDAR Point-Cloud Processing & Building Extraction** Coursework — Photogrammetry & Laser Scanning, Denmark
 - Processed raw airborne LiDAR (.las) tiles in CloudCompare — merging flight lines, noise filtering (SOR), and CSF ground/off-ground classification.
 - Built elevation, RGB and intensity surfaces in ArcGIS Pro and derived a raster DEM/DSM of the study area.
- National Atlas of the Sultanate of Oman** Team Cartography Project
 - Contributed to 25+ physical and human maps (terrain, geology, climate, population, services, infrastructure) as part of a student GIS team.
 - Applied thematic mapping (choropleth, graduated-symbol) in ArcGIS Pro to present geographic and demographic data.
- RUSLE Soil Erosion Assessment, North Sinai** NARSS Applied Study — July 2025
 - Cloud-based soil-erosion assessment using Google Earth Engine, Sentinel-2, CHIRPS rainfall and NDVI to compute RUSLE factors (R, K, LS, C, P).
 - Modelled before/after intervention scenarios, raising the 'slight erosion' class from ~38% to over 90% of the study area; assessed economic loss and priority zones.

CERTIFICATIONS & TRAINING

- Aerial Surveying (Photogrammetry) Course** — NARSS · February 2024
- RUSLE & GIS Cloud-Based Geospatial Analysis for Soil Erosion Assessment** — NARSS · July 2025
- Summer Field Training — Surveying** — Arab Contractors Co., Express Train Project · Evaluation: Excellent · August 2023

SKILLS

GIS Software ArcGIS Pro · ArcMap · ArcGIS Online · QGIS · ENVI · Google Earth Engine · CloudCompare · Global Mapper · Field Maps / Survey123	Programming & Tools Python (ArcPy) · SQL · ModelBuilder · Arcade · Git/GitHub · VS Code · MS Excel
Core GIS Skills Geodatabase management · Geoprocessing & spatial join · Topology · Raster/vector processing · 3D GIS & point-cloud processing · Web GIS	Spatial Analysis AHP & Weighted Overlay · RUSLE modelling · Kernel Density · Network Analyst · Gap Analysis · Suitability modelling
Remote Sensing & Surveying Sentinel-2 · Landsat-8 · SRTM DEM · CHIRPS rainfall · NDVI/NDBI/LST · Photogrammetry · LiDAR classification	Languages Arabic (Native) · English (Intermediate)